

KPI REQUIREMENTS

Interns must calculate the following KPIs:

Population KPIs

1. Total Penguins Count

- Total number of valid records after cleaning.

2. Species Distribution

- Count of penguins by species
- Percentage contribution by species

3. Island-wise Population

- Total penguins per island
- % distribution by island

Biological Measurement KPIs

4. Average Body Mass (g)

- Overall average
- Average by species
- Average by gender

5. Average Flipper Length (mm)

- Overall average
- By species
- By gender

6. Average Bill Length (mm)

- Overall average
- By species

7. Average Bill Depth (mm)

- Overall average
- By species

8. Bill Ratio KPI

- Bill Length / Bill Depth
- Average ratio by species

Gender-Based KPIs

9. Male vs Female Count

- Total males
- Total females
- Percentage split

10. Size Difference by Gender

- Avg body mass difference (Male vs Female)
- Avg flipper length difference

Correlation KPIs

11. Correlation Matrix

- Body mass vs flipper length
- Bill length vs bill depth

12. Highest Positive Correlation Pair

- Identify strongest relationship among numeric features

Island-Level KPIs

13. Species per Island

- Species count breakdown by island

14. Average Body Mass per Island

15. Average Flipper Length per Island

Data Quality KPIs

16. Missing Value Percentage

- Per column

17. Outlier Detection (Optional Advanced)

- Body mass outliers
- Flipper length outliers

CHART REQUIREMENTS

Interns must build the following visualizations:

Distribution Charts

- 1. Species Distribution (Bar Chart)**
 - X-axis: Species
 - Y-axis: Count
- 2. Island Distribution (Bar Chart)**
- 3. Gender Distribution (Pie or Donut Chart)**

Measurement Distribution Charts

- 4. Body Mass Distribution (Histogram)**
- 5. Flipper Length Distribution (Histogram)**
- 6. Bill Length Distribution (Histogram)**

Comparison Charts

- 7. Average Body Mass by Species (Bar Chart)**
- 8. Average Flipper Length by Species (Bar Chart)**
- 9. Average Body Mass by Gender (Bar Chart)**
- 10. Species Distribution by Island (Stacked Bar Chart)**

Relationship Charts

- 11. Flipper Length vs Body Mass (Scatter Plot)**

- Color by species
- Size by bill length (optional)

12. Bill Length vs Bill Depth (Scatter Plot)

13. Correlation Heatmap

Executive Dashboard Requirements

Dashboard must include:

Filters (Real-Time)

- Species (Multi-select)
- Island (Multi-select)
- Gender (Multi-select)

KPI Cards (Top Section)

- Total Penguins
- Avg Body Mass
- Avg Flipper Length
- Avg Bill Length
- Male vs Female %

Charts Section

- 1 Distribution chart
- 1 Comparison chart
- 1 Relationship chart
- 1 Correlation heatmap